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ISGF 2026 Theme Presentation

AMI → NextGenAMI

From Smart Metering to AI-Native Utilities

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Panel: AMI 2.0 – Grid Intelligence, DER Integration & Cyber-Resilient Architecture

The Real Question

We must digitize the grid — not just the billing process

Traditional AMI

- Programs focus strictly on remote reads, monthly monthly billing, and cash collection
- Limits data utility to simple cash-register functions functions and basic outage alerts

vs

AI-Native Utilities

- Transforms meters into a dense, real-time distributed sensor network
- Unlocks real-time grid observability and advanced consumer energy intelligence
- True value lies in the continuous intelligence extracted from the data stream

Treat meter data as a live operational signal, not a monthly input.

Utilities often operate without dependable system truth

Incorrect mappings: Consumer-to-transformer assignments are frequently wrong.

Outdated topology: Feeder and phase connectivity records records lag behind field changes.

Siloed systems: GIS, CIS, and MDM repositories erode cross-platform accuracy.

Operational Consequences

- Incorrect outage notifications
- Distorted commercial loss calculations
- Poor capacity & network planning
- Unreliable grid analytics & AI models

When system records drift from physical reality, every operational decision degrades.

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Same AMI data unlocks grid and consumer intelligence intelligence

Grid Intelligence

Use Case 1: Source of Supply Determination

- AI models infer which transformer actually supplies each consumer.
- Corrects outdated GIS/CIS records to build a true physical digital twin.
- Improves outage localization, asset planning, and loss calculations.

+

Consumer Intelligence

Use Case 2: Energy Disaggregation As A Foundation

- Detect appliance-level energy usage patterns directly from meter signals.
- Identifies shiftable load and major energy drains (e.g., HVAC, EVs).
- Enables highly targeted demand response and personalized consumer insights.

The value of AMI is a dual-intelligence platform, not a single-purpose monthly feed.

Pilot Insight 1

Meter data can reconstruct the grid's true topology

Pilot Scope

~53,000

Consumer meters analyzed

~1,000

Distribution transformers

67%

Consumers successfully mapped to their **actual supplying transformer** using analytics.

24%

Only 24% of existing **system records** matched matched the true physical connectivity.

27% DT assignments violated distance constraints.

22% Meters attributed to the wrong upstream feeder.

86%

Of corrected mappings fell within a highly reliable electrical distance band **(<300 m)**.

AMI analytics can continuously rebuild the true electrical topology of the grid.

Pilot Insight 2

AMI reveals how energy is used — not just how much

Pilot Scope

~39,000

Residential meters analyzed

4 Weeks

Analysis period for load profiles

763k

kWh of shiftable load identified across the analyzed residential fleet.

12-18 MW

Potential **peak demand reduction** through through targeted grid interventions.

10%

Of homes hold **~50%** of the total shiftable demand.

Alert

Meters marked inactive still drawing measurable power.

>220k

kWh per week of standby loads detected across the analyzed fleet.

Disaggregation turns billing data into actionable demand-side intelligence.

Real-World Outcomes

Making grid intelligence human and outcome-focused

Consumers

- Clearer explanations of high bills
- Personalized energy savings insights
- Better, more proactive service experience

Field Crews

- Accurate source-of-supply visibility
- Faster, more precise fault localization
- Reduced patrol time and quicker restorations

Planners

- Reliable, real-time transformer loading insights
- Early EV-adoption and DER readiness signals
- Highly accurate loss analytics and mapping

The Grid

- Targeted, high-yield demand response
- Significantly reduced peak stress on local assets
- Smarter, data-driven capacity investments

Takeaway: The best grid intelligence does not feel technical. It feels like better service.

Reference Architecture

AI-Native Utility Reference Architecture

Grid Edge Assets

- Consumers
- Electric Vehicles
- Distributed Solar
- Smart Devices

Smart Meter Edge

Embedded Analytics Node

- High-res event detection
- Local anomaly detection
- Appliance signature identification
- Edge-level filtering

Utility Data + AI Platform

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- Topology & connectivity inference
- Load disaggregation AI models
- DER visibility & EV detection
- Advanced loss analytics
- Cyber anomaly detection

Utility Systems

Operational Execution

- Grid Operations (ADMS)
- Outage Management (OMS)
- Network Planning Systems
- Customer Engagement
- DER Orchestration (DERMS)

Takeaway: NextGenAMI is not simply an infrastructure upgrade. It is the sensing layer of an AI-native utility architecture.

Leapfrog Opportunity

India can leapfrog by deploying AMI in the AI era

Early Global Adopters

- Deployed 15–20 years ago, before large-scale AI, EV adoption, and cloud economics.
- Architectures fundamentally optimized for automated monthly billing, not grid intelligence.
- Currently facing costly retrofits to handle modern decentralized grid complexities.

India's Leapfrog Advantage

- **AI-native analytics** embedded from Day One, not added as an afterthought.
- **Interoperable data architectures** taking full advantage of modern cloud scaling.
- **DER-ready grid intelligence** purposely designed for EVs and distributed solar.
- Creates a massive, scalable **distributed sensing network** straight straight out of the box.

Takeaway: Late adoption becomes a massive advantage when we design for where the grid is going, not where it was.

Closing Thought

Smart meters are often described as the last mile of the power system. But in reality, they may become the **nervous system of the grid.**

If AMI is used only for billing automation, we **digitize a process**.

vs

If AMI becomes a learning layer for the grid, we **change the operating model**.

When millions of smart meters begin speaking, the real question is:
Are utilities ready to learn from what the grid is telling them?

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India's late AMI adoption is a strategic advantage

- Early AMI worldwide optimized billing, not intelligence
- Today's grid is DER- and EV-heavy with AI everywhere
- India can design for the future grid from day one
- Use AMI to build a learning distribution grid, not just smarter meters

Late adoption enables an AI-native AMI foundation.

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